

Project 6: MCP-Powered Cross-Function Research Orchestration with Agent-to-Agent Collaboration

A C5i-style enterprise AI project concept for research and consulting operations.

Field	Details
Project	MCP-Powered Cross-Function Research Orchestration with Agent-to-Agent Collaboration
Business Themes	Enterprise AI; Research & Consulting Operations; AI-Powered Research Operations; Cross-functional workflow automation; Agent5i-style enterprise agent orchestration
Technology Stack / Themes	MCP (Model Context Protocol), Agent-to-Agent (A2A) architecture, FastMCP servers, tool registry, role-based access, multi-agent workflow, audit logs, orchestration layer
Goal	Implement a cross-function research orchestration demo where different departments expose tools through MCP servers. A user asks for a client-ready research update; the orchestrator coordinates Field/Collection, Analytics, and Client Servicing agents while respecting data access boundaries, tool permissions, and auditability.
Data / Inputs	Synthetic project plans, survey collection status, data quality reports, analysis tables, insight summaries, client emails, deliverable tracker, role-permission matrix
Evaluation Metrics	Correct tool selection, successful cross-agent task completion, permission enforcement, trace/audit completeness, PII-safe responses, orchestration latency, user satisfaction

High-Level Work Areas

- 1 Define three functional domains: Field/Collection, Analytics, and Client Servicing.
- 2 Build mock FastMCP servers for survey status, data quality, analysis outputs, project timeline, and client deliverables.
- 3 Create agents that discover tools from MCP servers instead of hardcoding every function.
- 4 Implement A2A message flow where agents collaborate on a client request.
- 5 Add role-based access and PII-safe responses.
- 6 Add audit trail for tool calls, inputs, outputs, and final recommendations.
- 7 Demonstrate how MCP enables modular tool integration and how A2A enables cross-functional collaboration.

Progressive Demo Sequence

- 1 Start with the client-servicing question: 'Prepare a status update for the client: field progress, data-quality issues, early insights, and risks.'
- 2 Show a monolithic agent with hardcoded functions. Highlight that adding/removing tools requires code changes and access control is weak.
- 3 Add MCP server discovery: Field MCP exposes survey progress, Analytics MCP exposes data-quality and insight tools, Client Servicing MCP exposes deliverable/timeline tools.
- 4 Ask the orchestrator: 'Which tools are available and which can this user access?' Show tool registry and permission filtering.
- 5 Execute progressive tool calls: get field completion, check sample gaps, fetch data-quality exceptions, retrieve early insights, and summarize timeline risk.
- 6 Add A2A collaboration: Field Agent explains collection gaps, Analytics Agent explains insight confidence, Client Servicing Agent drafts client message.
- 7 Add policy layer: block PII, mask respondent-level data, and prevent unauthorized raw-data export.
- 8 Add audit trail view: every tool call, agent handoff, evidence item, and final claim is logged.
- 9 Add failure scenario: one MCP server unavailable or returns stale data. Show fallback and final response with caveat.
- 10 Final limitations to highlight: MCP standardization is evolving, tool schemas need governance, multi-agent coordination can be brittle, and security testing is essential.
- 11 Future scope: enterprise SSO/RBAC, production tool registry, workflow memory, retry policies, observability, red-team testing, and integration with Agent5i-style enterprise agent workflows.

Conceptual Architecture

Layer	Responsibility	Example Components
User / Business Request	Client-servicing user asks for a research status update.	Request: field progress, data quality, early insights, risks
Orchestration Layer	Plans tasks, discovers tools, checks permissions, routes work across agents.	Orchestrator agent, planner, RBAC, audit logger
MCP Tool Layer	Exposes tools from different business functions in a modular way.	Field MCP, Analytics MCP, Client Servicing MCP
A2A Collaboration Layer	Enables specialized agents to exchange evidence, caveats, and outputs.	Field Agent, Analytics Agent, Client Servicing Agent
Governance Layer	Ensures PII safety, permission enforcement, traceability, and caveats.	PII masking, access policy, audit logs, stale-data checks
Output Layer	Produces a client-ready status update with evidence and risks.	Executive brief, evidence table, recommended next actions

Suggested Demo Questions

- Which tools are available for this user, and which tools are restricted?
- What is the latest field completion by segment and geography?
- Are there any data-quality issues or sample gaps that should be disclosed?
- What early insights can be shared with high confidence?
- What timeline risks should the client be warned about?
- Can you draft a client-ready status update with caveats and next steps?
- Show the audit trail for every tool call and agent handoff used in this answer.